1 Solution — GIAM 2.1 Problem 4

1.1 Problem

Complete truth tables for each of the sentences $(A \wedge B) \vee C$ and $A \wedge (B \vee C)$. Does it seem that these sentences have the same logical content?

1.2 Answer

$\boxed{\textbf{No}\text{—the two sentences are } \textit{not} \text{ logically equivalent.}}$

They agree on 6 of the 8 input rows but disagree on the two rows where $A = 0$ and $C = 1$.

1.3 The Truth Tables

Building both formulas column by column over all $2^3 = 8$ combinations of A, B, C:

Both formulas over all 8 rows. The two final columns disagree in the shaded rows — so they are not equivalent.

A	B	C	A∧B	(A∧B)∨C	B∨C	A∧(B∨C)
0	0	0	0	0	0	0
0	0	1	0	**1**	1	**0**
0	1	0	0	0	1	0
0	1	1	0	**1**	1	**0**
1	0	0	0	0	0	0
1	0	1	0	1	1	1
1	1	0	1	1	1	1
1	1	1	1	1	1	1

Rows 2 and 4 differ: when A = 0 and C = 1, the left side is 1 (from C) but the right side is 0 (needs A).

Same output in 6 of 8 rows, but different in 2 ⟹ **NOT logically equivalent.**

The parentheses genuinely matter: "and" and "or" do not regroup freely across each other.

Figure 1.1: A combined truth table with columns A, B, C, A and B, (A and B) or C, B or C, and A and (B or C), over all 8 rows. The two output columns (A and B) or C and A and (B or C) are equal in six rows but differ in two shaded rows: (A,B,C) = (0,0,1) where left is 1 and right is 0, and (A,B,C) = (0,1,1) where left is 1 and right is 0. In both differing rows A = 0 and C = 1, so the left side is true because of C alone while the right side is false because it requires A. Conclusion: not logically equivalent.

A	B	C	$A \wedge B$	$(A \wedge B) \vee C$	$B \vee C$	$A \wedge (B \vee C)$
0	0	0	0	0	0	0
0	0	1	0	**1**	1	**0**
0	1	0	0	0	1	0
0	1	1	0	**1**	1	**0**
1	0	0	0	0	0	0
1	0	1	0	1	1	1
1	1	0	1	1	1	1
1	1	1	1	1	1	1

Table 1.1

The two boldfaced rows are where the final columns disagree.

1.4 Where and Why They Differ

The formulas part ways exactly when $A = 0$ and $C = 1$ (rows $(0,0,1)$ and $(0,1,1)$):

- $(A \wedge B) \vee C$ is *true* there, because the lone disjunct C is true — and an OR needs only one true piece.
- $A \wedge (B \vee C)$ is *false* there, because the outer AND demands A, and $A = 0$ kills the whole conjunction regardless of B or C.

So the disagreement is entirely about whether a true C can "carry" the sentence on its own. In the first formula it can; in the second it cannot, because C is trapped inside a conjunction with A.

1.5 Remark — Parentheses Are Not Decorative

The only difference between the two sentences is *where the parentheses sit*, and that difference changes the meaning. This is the logical analogue of arithmetic's $(2 \times 3) + 4 = 10$ versus $2 \times (3 + 4) = 14$: regrouping is not free. The lesson the exercise drives home is that $\wedge$ and $\vee$ do **not** associate *with each other* — you may not slide parentheses across a mixed $\wedge/\vee$ expression. (They *do* each associate with themselves: $(A \wedge B) \wedge C = A \wedge (B \wedge C)$, and likewise for $\vee$. It is only the mixing that breaks.) Whenever both connectives appear, the grouping must be written explicitly or the sentence is genuinely ambiguous.

1.6 Remark — The Correct Distributive Law

There *is* a valid way to "multiply out" a mixed expression — the **distributive law** — but it introduces a repeated variable rather than simply moving a parenthesis:

$$A \wedge (B \vee C) \;=\; (A \wedge B) \vee (A \wedge C).$$

Notice the A appears *twice* on the right. The naive $(A \wedge B) \vee C$ of this problem is missing that second copy of A (it has a bare C instead of $A \wedge C$), which is precisely why it comes out different. Distribution is the analogue of $a \times (b + c) = ab + ac$ in arithmetic — and just as there, you cannot drop one of the two products. The exercise is a concrete warning against the mistake of "distributing" by only relocating a parenthesis.

1.7 Remark — When Do the Two *Happen* to Agree?

Although not equivalent in general, the two formulas coincide on 6 of 8 rows, and it is worth seeing which. They agree whenever $A = 1$ (all four such rows: then both reduce to $B \vee C$ vs. 1... let us check — with $A = 1$, left is $B \vee C$ and right is $B \vee C$, identical), and also on the rows with $C = 0$ (then left is $A \wedge B$ and right is $A \wedge B$, identical). The *only* way to break them is to make the bare C decisive: C true while A false. So the equivalence fails on exactly the two rows where those conditions coincide.

Spotting *when* two non-equivalent formulas nonetheless match is a useful habit — it often reveals the extra hypothesis under which a tempting-but-false identity becomes true (here: “if A is known true, the two sentences *are* interchangeable”).

1.8 Remark — Truth Tables as a Decision Procedure

This problem showcases the truth table as a complete, mechanical **decision procedure** for equivalence: two sentences have the same logical content *if and only if* their final columns match in every row, and a single differing row disproves equivalence outright. No cleverness is required — just fill in all 2^n rows for the n variables. The cost is that the table doubles with each new variable (2^n rows), which is why, for larger formulas, logicians turn to the algebraic laws (De Morgan, distribution, association) that let one *transform* one sentence into another symbolically. But as a guaranteed-to-terminate check, the truth table is the bedrock — and here it settles the question in eight lines.